

Unique Paper Code : 32341401
Name of the Course : B. Sc. (Hons.) Computer Science – CBCS (old course)
Name of the Paper : Design and Analysis of Algorithms
Semester : IV
Year of admission : 2015,2016,2017 and 2018
Duration : 3 Hours
Maximum Marks : 75

Instructions for Candidates:

Attempt Any Four questions. All Questions carry equal marks.

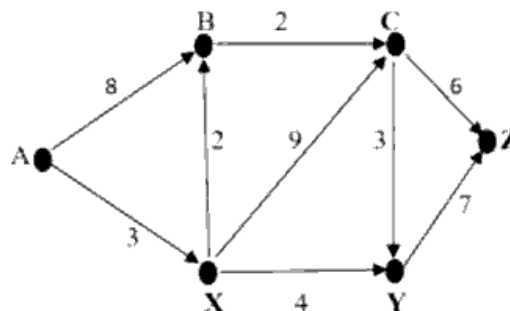
1. A student named Rohan wishes to enhance his knowledge by studying a few online courses. Every course has a fee that one has to pay to register. Also, each course gives some credit on its successful completion. He has some amount of money saved with him. He needs to choose among the available courses such that he gets maximum sum of credits by spending the money he has. Write an algorithm to find the optimal solution for Rohan's problem.

Using the algorithm given by you, find the courses he should study for the given instance, illustrating each step clearly.

Rohan has \$70.

S. No.	Course Name	Fee (in \$)	Credit
1.	C1	40	12
2.	C2	20	8
3.	C3	30	16
4.	C4	10	9
5.	C5	30	5

2. A person named **Albert (denoted by A)** starts a food delivery service. The people who wish to avail this service register themselves with him by providing their address. He hires delivery boys to deliver food and assigns a different delivery boy to each registered person. To save time, he asks the delivery boys to use Scooters to deliver the food. The following diagram shows the time that a delivery boy takes to reach from one place to another. Consider that **B, C, X, Y** and **Z** are the registered people. The number on each edge represents the time (in minutes). Compute the **minimum** time to reach them all individually.



Now, suppose all Scooters need to be serviced and are unavailable. The delivery boys need to use bicycles to reach the registered people. The amount of time taken to reach the destination using a bicycle is the square of the time taken to reach the destination using a

Scooter. Will the delivery boys be reaching the respective destination using same route as computed above? Does this imply all such routes will never change in any such problem instance where the new edge weights are square of original edge weights?

3. Suppose there are n bottles and each bottle i has some capacity c_i . Write the most efficient algorithm to arrange the bottles in the order of their capacities. Also, explain the running time of your algorithm.

Next, consider that every bottle has an expiry date before which its content should be consumed. Each bottle is assigned a priority such that a higher value indicates a higher priority. The bottle having a closer expiry date has a higher priority assigned to it. Give an efficient algorithm to arrange the bottles in non-increasing order of their priorities. Does your algorithm involve any comparisons? If yes, count them. Explain the running time of your algorithm.

4. Consider a set of jobs to be scheduled on a processor. Each job has an execution time associated with it. A job-scheduler allocates the processor to one of the given jobs at a time. The scheduler has fixed a maximum execution time: the maximum duration for which a job can continue processing once scheduled. If the execution time of any job is greater than this value then it is pre-empted and scheduled again later with the remaining execution time.

Consider the following variant of **shortest-job-first** scheduling algorithm which schedules the jobs in the ascending order of their execution times. Suppose the data structure used for this scheduling algorithm is a red-black tree (called as scheduling tree) where the jobs are the nodes in the tree.

Algorithm:

- The job having least execution time will be chosen and sent for execution and is removed from the tree.
- If the job reaches its maximum execution time while being processed, it is pre-empted and needs to be scheduled later to complete its processing. This job is reinserted into the scheduling tree based on its remaining execution time.
- Again, the job with the least execution time is selected, repeating the above steps until all the jobs are processed completely.

Consider a set of jobs whose execution times (in nanoseconds) are as follows: **4,10,7,23,5,30**. Construct the red-black tree corresponding to the given input and then schedule the jobs using the above stated algorithm thereby showing the red-black tree after every step. Assume **the maximum execution time is 20 nanoseconds**.

5. A party is going on in a hotel. When the dinner starts, a waiter is assigned the task of distributing plates to the guests. He creates a pile of plates on a table by adding one plate at a time. The waiter can pick up one or more plates at a time from the pile to distribute. The waiter may add more plates to the pile one by one at any time. Suppose he performs a total of n operations wherein an operation can be putting one plate onto the pile or removing one or more plates from the pile. Analyse the average cost per operation over a sequence of n operations.

Later, after the dinner gets over, a game is played to find out the most social person in the party. Suppose there are m guests in the party. Each guest considers a set of guests as his friends and gives a flower to each of his friends. The guest who receives the maximum number of flowers wins the game. Suggest an efficient divide-and-conquer algorithm, having least possible worst-case running time, to find a winner. Also, find all the winners, if more than one winner exists. Explain the running of the algorithm given by you.

6. A chemical laboratory has n chemicals stored in it. There is a risk that some chemicals may react with some others. For safety purpose, the lab assistant decides that for each pair of reactive chemicals, one is kept in a yellow-coloured bottle and the other is kept in a green-coloured bottle. Give an efficient algorithm that he can use to find out whether such a colouring is possible or not. If yes, the algorithm should determine the bottle colour for each of the n chemicals. If not, the algorithm should report it. Give an instance having at least 6 chemicals in which there are a minimum of 7 reactive pairs for which such a colouring is possible.